Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In the reaction

$3Br_2 + 6CO_3^{2-} + 3H_2O \longrightarrow$

$$5Br^- + BrO_3^- + 6HCO_3^-$$

(A) bromine is oxidised and carbonate is reduced.

(B) bromine is reduced and water is oxidised

(C) bromine is neither reduced nor oxidised

(D) bromine is both reduced and oxidised

Q.2

Which one of the following cannot function as an oxidising agent?

(A) I^-

(B) S(s)

(C) $NO_3^-(aq)$

(D) $Cr_2O_7^{2-}$

Q.3

Which of the following chemical reactions depict the oxidizing beahviour of H_2SO_4 ?

(A) $NaCl + H_2SO_4 \longrightarrow NaHSO_4 + HCl$

(B) $2PCl_5 + H_2SO_4 \longrightarrow 2POCl_3 + 2HCl + SO_2Cl_2$

(C) $2HI + H_2SO_4 \longrightarrow I_2 + SO_2 + 2H_2O$

(D) $Ca(OH)_2 + H_2SO_4 \longrightarrow CaSO_4 + 2H_2O$

Q.4

Consider the reaction:

$$H_2SO_3(aq) + Sn^{4+}(aq) + H_2O(l) \rightarrow Sn^{2+}(aq) + HSO_4^-(aq) + 3H^+(aq)$$

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Which of the following statements is correct?

(A) Sn^{4+} is the oxidizing agent because it undergoes oxidation

(B) Sn^{4+} is the reducing agent because it undergoes oxidation

(C) H_2SO_3 is the reducing agent because it undergoes oxidation

(D) H_2SO_3 is the reducing agent because it undergoes reduction

Q.5

Which substance serves as reducing agent in the following reaction ?

$$14H^+ + Cr_2O_7^{2-} + 3Ni \longrightarrow 2Cr^{3+} + 7H_2O + 3Ni^{2+}$$

(A) H_2O

(B) Ni

(C) H^+

(D) $Cr_2O_7^{2-}$

Q.6

Thiosulphate reacts differently with iodine and bromine in the reactions given below.

$$2\,S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$$

$$S_2O_3^{2-} + 2Br_2 + 5H_2O \rightarrow 2SO_4^{2-} + 2Br^- + 10H^+$$

Which of the following statements justifies the above dual behaviour of thiosulphate?

(A) Bromine is a stronger oxidant than iodine

(B) Bromine is a weaker oxidant than iodine

(C) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reactions

(D) Bromine undergoes oxidation and iodine undergoes reduction in these reactions

Q.7

Which reaction involves neither oxidation nor reduction?

(A) $CrO_4^{2-} \longrightarrow Cr_2O_7^{2-}$

(B) $Cr \longrightarrow CrCl_3$

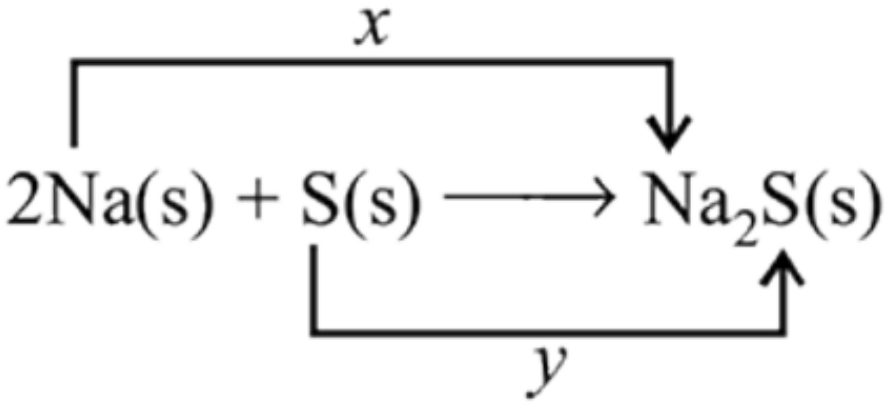
(C) $Na \longrightarrow Na^+$

(D) $2\,S_2O_3^{2-} \longrightarrow S_4O_6^{2-}$

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Q.8

Which of the following is correct code for x and y in the following reaction.



(i) x = oxidation reaction, y = reduction reaction
(ii) x = gain of two electrons, y = loss of two electrons,
(iii) x = reduction reaction, y = oxidation reaction
(iv) x = loss of two electrons, y = gain of two electrons

(A) (i) and (ii)

(B) (i) and (iv)

(C) (ii) and (iii)

(D) (iii) and (iv)

Q.9

One gas bleaches the colour of flowers by reduction, while the other by oxidation

(A) CO and Cl_2

(B) SO_2 and Cl_2

(C) H_2 S and Br_2

(D) NH_3 and SO_2

Q.10

In the reaction :
$C + 4HNO_3 \rightarrow CO_2 + 2H_2O + 4NO_2$
HNO_3 act as

(A) an oxidizing agent

(B) an acid

(C) an acid as well as oxidizing agent

(D) a reducing agent

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Q.11

The standard reduction potentials at $298\ \mathrm{K}$ for the following half reactions are given against each
$\mathrm{Zn}^{2+}(\mathrm{aq}) + 2\mathrm{e} \rightleftharpoons \mathrm{Zn}(\mathrm{s}); -0.762\ \mathrm{V}$
$\mathrm{Cr}^{3+}(\mathrm{aq}) + 3\mathrm{e} \rightleftharpoons \mathrm{Cr}(\mathrm{s}); -0.740\ \mathrm{V}$
$2\mathrm{H}^{+}(\mathrm{aq}) + 2\mathrm{e} \rightleftharpoons \mathrm{H}_2(\mathrm{g}); 0.00\ \mathrm{V}$
$\mathrm{Fe}^{3+}(\mathrm{aq}) + \mathrm{e} \rightleftharpoons \mathrm{Fe}^{2+}(\mathrm{aq}); 0.770\ \mathrm{V}$
Which is the strongest reducing agent?

(A) $\mathrm{Zn}(\mathrm{s})$

(B) $\mathrm{Cr}(\mathrm{s})$

(C) $\mathrm{H}_2(\mathrm{g})$

(D) $\mathrm{Fe}^{3+}(\mathrm{aq})$

Q.12

In the reaction shown below, oxidation state of the carbon in reactant and product are (i) and (ii) respectively? Is the given reaction a redox reaction? $\mathrm{Na_2CO_3(aq)} + \mathrm{HCl(aq)} \longrightarrow \mathrm{Na^+(aq)} + \mathrm{Cl^-(aq)} + \mathrm{H_2O(l)} + \mathrm{CO_2(g)}$

(A) (i) 6, (ii) 4 , yes

(B) (i) 6, (ii) $6, \mathrm{No}$

(C) (i) 4, (ii) 4, No

(D) (i) 4, (ii) 4, yes

Q.13

In how many of the following species, an element has fractional oxidation state?
i. $\mathrm{C_3O_2}$
ii. $\mathrm{Fe_3O_4}$
iii. $\mathrm{CrO_2Cl_2}$
iv. $\mathrm{Fe_2O_3}$
v. $\mathrm{Cr_2O_3}$
vi. $\mathrm{Br_3O_8}$
vii. $\mathrm{Na_2S_4O_6}$
viii. $\mathrm{Pb_3O_4}$

(A) 5

(B) 3

(C) 7

(D) 6

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Q.14

On addition of I^- with MnO_4^- in presence of acidic medium, it convert into

(A) IO_3^-

(B) IO_4^-

(C) I_2

(D) IO_2^-

Q.15

The most powerful reducing agent is:

(A) H_3PO_3

(B) H_3PO_4

(C) $H_4P_2O_7$

(D) $H_4P_2O_6$

Q.16

Following order is observed in oxidising power of certain ions:
$VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$
The reason for this increasing order of oxidising power is:

(A) Increasing stability of the lower species to which they are reduced.

(B) Increasing stability of the higher species to which they are oxidised.

(C) Increasing stability of the higher species to which they are reduced.

(D) Increasing stability of the lower species to which they are oxidised.

Q.17

$NH_3 + OCl^{\ominus} \longrightarrow N_2H_4 + Cl^{\ominus}$ On balancing the above equation in basic solution, using integral coefficient, which of the following whole numbers will be the coefficient of N_2H_4?

(A) 1

(B) 2

(C) 3

(D) 4

Q.18

In Na_2O_2, the oxidation states respectively for Na and oxygen may be:

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(A) $+1, \ -1$

(B) $+2, \ -2$

(C) $+1, \ -2$

(D) None of these

Q.19

Which of the following is used as a good oxidising agent in analytical chemistry ?

(A) $\mathrm{Gd(III)}$

(B) $\mathrm{Ce(IV)}$

(C) $\mathrm{Eu(II)}$

(D) $\mathrm{Lu(III)}$

Q.20

Oxidation number of potassium in K_2O, K_2O_2 and KO_2, respectively:

(A) $+1, \ +1, \ +1$

(B) $+\frac{1}{2}, \ +4, \ +1$

(C) $+2, \ +1, \ +\frac{1}{2}$

(D) $+\frac{1}{2}, \ +1, \ +2$

Answers & Solutions

Q.1 Answer:

bromine is both reduced and oxidised

Solution:

$3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow$
$5Br^- + BrO_3^- + 6HCO_3^-$

O.N. of Br_2 changes from 0 to -1 and $+5$ hence it is reduced as well as oxidised.

Q.2 Answer:

I^-

Solution:

If an electronegative element is in its lowest possible oxidation state in a compound or in free state. It can function as a powerful reducing agent. e.g. I-

Q.3 Answer:

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$2HI + H_2SO_4 \longrightarrow I_2 + SO_2 + 2H_2O$

Solution:

$2HI^{-1} + H_2^{+6}SO_4 \longrightarrow I_2^0 + \overset{+4}{S}O_2 + 2H_2O$

In thisreaction oxidation number of S is decreasing from $+6$ to $+4$ hence undergoing reduction and for HI oxidation Number of I is increasing from -1 to 0 hence underegoing oxidation therefore H_2SO_4 is acting as oxidising agent.

Q.4 Answer:

H_2SO_3 is the reducing agent because it undergoes oxidation

Solution:

$$H_2SO_3(aq) + Sn^{4+}(aq) + H_2O(l) \underset{+6}{\longrightarrow} Sn^{2+}(aq) + HSO_4^-(aq) + 3H^+$$

Hence H_2SO_3 is the reducing agent because it undergoes oxidation.

Q.5 Answer:

Ni

Solution:

The compound undergo oxidation itself and reduces others is known as reducing agent. In this reaction O.

N. of Ni changes from 0 to $+2$ and hence Ni acts as a reducing agent.

Q.6 Answer:

Bromine is a stronger oxidant than iodine

Solution:

$2\ S_2O_3^{2-}(aq) + I_2(\ s) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

$^{+2}\ S_2O_3^{2-}(aq) + 2Br_2(l) + 5H_2O(l) \rightarrow$

$2SO_4^{2-}(aq) + 4Br^-(aq) + 10H^+(aq)$

Hence, bromine is a stronger oxidising agent than I_2,

as it oxidises S of $S_2O_3^{2-}$ to SO_4^{2-} whereas I_2 oxidises

it only into $S_4O_6^{2-}$ ion.

Q.7 Answer:

$CrO_4^{2-} \longrightarrow Cr_2O_7^{2-}$

Solution:

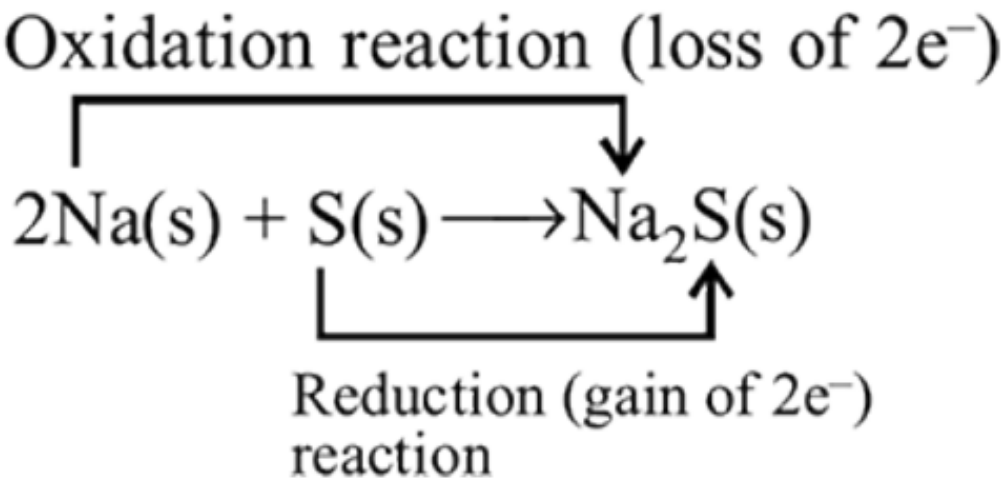
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O. N. of Cr on both sides is +6.

Q.8 Answer:

(i) and (iv)

Solution:



Q.9 Answer:

SO_2 and Cl_2

Solution:

SO_2 bleaches by reduction, while chlorine bleaches colour of flowers by oxidation.

Q.10 Answer:

an oxidizing agent

Solution:

O.N. of C changes form 0 to +4 by oxidation. Hence HNO_3 is oxidising agent.

Q.11 Answer:

Zn(s)

Solution:

Since oxidation potential of Zn is highest hence strongest reducing agent.

Q.12 Answer:

(i) 4, (ii) 4, No

Solution:

The redox reaction involve loss or gain of electron(s) i.e. change in oxidation state. Given reaction is

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not a redox reaction as this reaction involves no change in oxidation state of reactant or product.

Q.13 Answer:

5

Solution:

In species *i, ii, vi, vii* **and** *viii,* **an element has fractional oxidation state.**

Q.14 Answer:

I_2

Solution:

In acidic medium, the iodide will oxidize to molecular iodine:

$$2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$$

Q.15 Answer:

H_3PO_3

Solution:

H_3PO_3 **oxidation state of P is+3 , which can further be oxidized to a higher oxidation state. So they can act as reducing agents. In** H_3PO_4 **oxidation state of P is +5, which is highest for P. Hence** H_3PO_4 **cannot serve as a reducing agent.**

Q.16 Answer:

Increasing stability of the lower species to which they are reduced.

Solution:

The given order is : $VO_2^+ < Cr_2O_7^{2-} < MnO_4^{2-}$

The oxidation states of the various transition elements are:

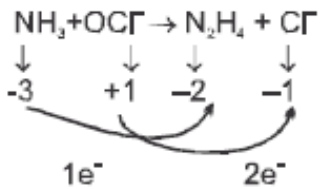
$V - +5; Cr - +6; Mn - +6$

We know that heavier members stabilises higher oxidation state in case of transition elements. So, the given order of oxidising power is based on increasing stability of lower species to which they are reduced.

Q.17 Answer:

1

Solution:



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$2NH_3 + OCl^- \longrightarrow N_2H_4 + Cl^- + H_2O$

Coefficient of $N_2H_4 = 1$

Q.18 Answer:

$+1, \quad -1$

Solution:

Let x **be the oxidation state of** Na **in** Na_2O_2 .

Na_2O_2 **contains a peroxide linkage in which** O **has an oxidation state of** -1 .

$\overset{x}{Na_2}\overset{-1}{O_2}$ **or** $2x + 2\ (-1) = 0$ **or** $x = +1$.

Thus, the oxidation state of sodium in Na_2O_2 **is** $+1$.

Q.19 Answer:

CeIV

Solution:

Ce^{4+} **has a tendency to act as an oxidizing agent as it is most stable in the** $+3$ **oxidation state.** Ce^4 **readily gains an electron to form** Ce^{3+}**. Thus, cerium is reduced and it oxidises another element.** Ce^{4+}**, thus, acts as an oxidising agent.**

Q.20 Answer:

$+1, \quad +1, \quad +1$

Solution:

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